Ishaan Handa

Purdue University, B.S. Computer Science | Expected December 2026 | Software Engineering, Machine Learning, and Cloud Systems
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SUMMARY

Computer Science and Machine Learning student at Purdue University with experience across full-stack development, distributed systems, and applied AI/ML, including agentic LLM systems, model fine-tuning, and cloud infrastructure on AWS and Azure. Founded and led an engineering team as CTO of a startup and interned across a global entertainment studio, a Fortune 25 company, national laboratories, and a data and analytics consultancy, building on coursework in operating systems, systems programming, data structures and algorithms, and object-oriented design.

SKILLS

Languages: Java, Python, C++, C, JavaScript, TypeScript, SQL

Web and Backend: React, Node.js, Express, REST APIs, event-driven architecture, schema design, authentication, HTML, CSS

Cloud and DevOps: AWS (Lambda, DynamoDB, EventBridge, S3, Cognito, CloudFormation, SQS, QuickSight, Redshift, Textract), Azure, HashiCorp Vault, Docker, Kubernetes, Terraform, CI/CD (GitLab)

AI and Machine Learning: LLMs (Bedrock, Gemini, Claude), agentic and multi-agent systems, MCP (Model Context Protocol), PydanticAI, tool-calling, prompt engineering (few-shot, chain-of-thought), model fine-tuning, TensorFlow, PyTorch, scikit-learn, NLP, vLLM

WORK EXPERIENCE

AI/ML Platform Engineering Intern, *Walt Disney Animation Studios* *May 2026 – Aug 2026 | Burbank, CA*

- Designed and built an agentic LLM system (Python, PydanticAI) that reduced security credential provisioning time from over 24 hours to under 5 minutes, using LLM classification with few-shot and chain-of-thought prompts to extract structured fields from unstructured Jira tickets, then autonomously executing YAML configuration and GitLab CI/CD tasks through tool-calling.
- Designed and deployed multiple independent MCP servers, exposed to the agent via streamable HTTP for on-demand tool access, avoiding the token overhead of loading full tool specifications into context. Built a classification evaluation suite to measure precision and recall on ground-truth datasets, using it to iteratively improve ticket triage accuracy.
- Engineered a fault-tolerant recovery system pairing semantic CI failure diagnosis with a SQLite-backed retry state machine, identifying root-cause YAML misconfigurations, triggering targeted git rollbacks, auto-retrying failed provisioning up to 3x, and notifying requesters on permanent failure with zero manual on-call intervention.
- Deployed the system on Kubernetes with a 5-minute polling schedule and 20 parallelized LLM inference workers, enabling continuous concurrent request throughput and eliminating all 504 Gateway Timeout errors under production load.

AI and Application Development Intern, *Centene Corporation* *May 2025 – Aug 2025 | St. Louis, MO*

- Developed iBills, an end-to-end AI-powered billing pipeline using AWS Bedrock LLMs to automate reconciliation and transform fragmented billing data into structured, actionable insights.
- Built distributed data pipelines with AWS Lambda, DynamoDB, and Textract, achieving a 5x improvement in processing accuracy and enabling real-time analytics.
- Engineered scalable backend services in Java and Python on AWS with object-oriented design, applying secure coding practices and architecting event-driven systems with EventBridge, SQS, Cognito authentication, and CloudFormation infrastructure as code.
- Selected as one of 6 Peer Leaders among 170+ interns and one of 7 interns invited to present technical work to executive IT leadership.

CTO and Founding Engineer, *Chainge STL* *Mar 2025 – Jan 2026 | St. Louis, MO*

- Founded and built a full-stack, consumer-facing civic engagement platform from scratch (React, Node.js, Express, REST APIs), owning architecture and product decisions end to end, deployed on Vercel with the Google Gemini API powering AI-generated event summaries.
- Recruited and led a team of 4 engineers, overseeing system design, code review, and testing to ensure high-quality, secure delivery.

Data Science Intern, *KNOWiNK* *May 2024 – Aug 2024 | St. Louis, MO*

- Engineered Python and SQL data pipelines to process large-scale, real-time election data, improving speed and reliability by 70 percent while enabling high-frequency ingestion.
- Built and optimized AWS QuickSight dashboards and Redshift queries, achieving a 30 percent runtime reduction for real-time KPI reporting.

Team Member, *Sandia National Labs* *Aug 2023 – Apr 2024 | West Lafayette, IN*

- Collaborated with Sandia National Labs to design, build, and test a Python-based flight prediction tool, applying pattern recognition and predictive modeling techniques to truncated flight path and commercial data to deliver reliable, safety-focused models.

PROJECTS

BriefCase: Fine-tuned DistilBERT and Legal-BERT on CUAD (13,000+ annotated legal clauses) using chain-of-thought rationales from Llama 3.1 8B, achieving 0.89 macro F1 on a 10-class contract risk classification task. Built and tested a fully on-premise inference pipeline using vLLM and PyTorch, and benchmarked it against direct LLM zero-shot inference, which it outperformed by roughly 47 macro F1 points.

Linux Shell: Developed a Unix-compatible shell in C with subshells, redirection, and job control, working directly with process management and system calls on Linux. Wrote tested, documented, and maintainable code following secure coding practices.

EDUCATION

Purdue University, B.S. Computer Science (Machine Learning and Software Engineering track)
Minors: Mathematics, Entrepreneurship | Expected Graduation: December 2026